Exercise Sheet

Lecture 7

Inner Product Spaces

The inner product is taken in the physics convention: conjugate-linear in the first slot, linear in the second, so $\langle \phi \mid \psi \rangle = \langle \phi, \psi \rangle$ is linear in the ket. Unless stated otherwise, the inner product is the standard one. Tags mark the flavour; a ⋆ marks a harder one.

Core tutorial route

Work **7.2, 7.1(i), 7.6, 7.8(i), 7.10, and 7.14(iii)** in that order. This route covers the inner-product axioms, computing in $\mathbb{C}^n$, the orthogonal decomposition, a proof of the Cauchy–Schwarz inequality in $\mathbb{C}^2$, the polarization identity, and a quantum amplitude.

Additional practice: 7.1(ii), 7.3–7.5, 7.7(i), 7.8(ii), 7.9, 7.11, 7.13, 7.16. **Bridge:** 7.1(iii), 7.7(ii), 7.15 (function spaces and a rank-one projector). **Extension:** 7.12 (the equality case of the triangle inequality), 7.14(i)–(ii) (expectation values).

Inner products, norms, and orthogonality

Exercise 7.1 (computational)

(i) In $\mathbb{C}^3$ take $u = (1, -i, 2)$ and $v = (i, 1, 1-i)$. Compute $\langle u, v\rangle$, $\langle v, u\rangle$, $\|u\|$, and $\|v\|$.

(ii) On $M_{2\times 2}(\mathbb{R})$ with the Frobenius inner product $\langle A, B\rangle = \operatorname{tr}(A^\dagger B)$, take $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 0 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Compute $\langle A, B\rangle$, $\|A\|$, and $\|B\|$.

(iii) On $\mathscr{C}[0,1]$ with $\langle f, g\rangle = \int_0^1 \overline{f} g \, dt$, take $f(t) = 1 + t$ and $g(t) = 1 - t$. Compute $\langle f, g\rangle$, $\|f\|$, and $\|g\|$.

Answer. (i) $\langle u, v\rangle = 2$, $\langle v, u\rangle = 2$, $\|u\| = \sqrt{6}$, $\|v\| = 2$. (ii) $\langle A, B\rangle = 4$, $\|A\| = 3$, $\|B\| = \sqrt{6}$. (iii) $\langle f, g\rangle = \frac{2}{3}$, $\|f\| = \sqrt{7/3}$, $\|g\| = \frac{1}{\sqrt{3}}$.

Exercise 7.2 (proof) On $\mathbb{C}^2$ consider the form $B(u, v) = \overline{u_1} v_1 - \overline{u_2} v_2$. Decide whether B is an inner product in the physics convention. Check conjugate symmetry, linearity in the second slot, and positive definiteness separately.

Answer. B is conjugate-symmetric and linear in the second slot, but is not positive definite: $B((0,1),(0,1)) = -1$.

Hint. Two of the axioms can be checked directly. For the third, look for a nonzero vector on which B takes a negative value; try e_2.

Exercise 7.3 (computational) Find all a such that the two vectors are orthogonal: (i) $(1, a, 2)$ and $(3, -1, 1)$ in $\mathbb{R}^3$, $a \in \mathbb{R}$; (ii) $(1, a)$ and $(1, i)$ in $\mathbb{C}^2$, $a \in \mathbb{C}$.

Answer. (i) $a = 5$. (ii) $a = -i$.

Exercise 7.4 (computational) In $\mathbb{R}^3$ take $v_1 = (1, 1, 0)$ and $v_2 = (0, 1, 1)$. Compute the Gram matrix $G_{jk} = \langle v_j, v_k\rangle$ of these vectors, and use it to decide whether the vectors are linearly independent.

Answer. $G = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\det G = 3 \neq 0$, so the vectors are linearly independent.

Exercise 7.5 (proof)

(i) Let U be a subspace of an inner product space V. Prove that $U \cap U^{\perp} = \{0\}$.

(ii) Let $v_1, \dots, v_m$ be nonzero, pairwise orthogonal vectors. Prove that they are linearly independent.

Hint. (i) A vector lying in both sets is orthogonal to every vector of U, itself included. (ii) Pair the equation $\sum_k c_k v_k = 0$ with each v_j; by orthogonality only one term survives.

The Cauchy–Schwarz inequality

Exercise 7.6 (computational) In $\mathbb{R}^2$ take $u = (4, 2)$ and $v = (1, 1)$. Split the vector u into a part parallel to v and a part orthogonal to v. Find the parallel part by the formula $\frac{\langle v, u\rangle}{\|v\|^2}\, v$.

Answer. $u = (3, 3) + (1, -1)$, where $(3, 3) = 3v$ is parallel to v and $(1, -1) \perp v$.

Exercise 7.7 (computational)

(i) In $\mathbb{C}^2$ take $u = (1, 1)$ and $v = (1, i)$. Compute $|\langle u, v\rangle|$, $\|u\|$, and $\|v\|$, and check the Cauchy–Schwarz inequality.

(ii) On $\mathscr{C}[0, 1]$ with $\langle f, g\rangle = \int_0^1 \overline{f}\, g\, dt$, compute $\langle 1, t^2\rangle$, $\|1\|$, and $\|t^2\|$, and check the Cauchy–Schwarz inequality for the functions 1 and t^2.

Answer. (i) $|\langle u, v\rangle| = \sqrt{2}$, $\|u\| = \|v\| = \sqrt{2}$; the Cauchy–Schwarz inequality $\sqrt{2} \le 2$ holds. (ii) $\langle 1, t^2\rangle = \frac{1}{3}$, $\|1\| = 1$, $\|t^2\| = \frac{1}{\sqrt{5}}$; the Cauchy–Schwarz inequality $\frac{1}{3} \le \frac{1}{\sqrt{5}}$ holds.

Exercise 7.8 (proof)

(i) For $u, v \in \mathbb{C}^2$ prove the identity $\|u\|^2\|v\|^2 - |\langle u, v\rangle|^2 = |u_1 v_2 - u_2 v_1|^2$. Deduce the Cauchy–Schwarz inequality in $\mathbb{C}^2$ and its equality case.

(ii) Use the Cauchy–Schwarz inequality to prove that for all $a_1, \dots, a_n \in \mathbb{C}$ we have $\left|\sum_{k=1}^n a_k\right|^2 \le n \sum_{k=1}^n |a_k|^2$.

Hint. (i) Expand both sides; the terms $|u_1|^2|v_1|^2$ and $|u_2|^2|v_2|^2$ cancel on the left. For the equality case, $u_1 v_2 - u_2 v_1$ is the determinant of the matrix with columns u, v. (ii) Apply the Cauchy–Schwarz inequality on $\mathbb{C}^n$ to $u = (1, 1, \dots, 1)$ and $v = (a_1, \dots, a_n)$, then read off $\|u\|^2$.

Triangle inequality, parallelogram law, and polarization

Exercise 7.9 (computational) Compute $\langle u, v\rangle$, $\|u\|$, $\|v\|$, and the cosine of the angle between the vectors u and v, where (i) $u = (1, 2, 2)$ and $v = (2, 2, 1)$ in $\mathbb{R}^3$; (ii) $u = (1, 1, 1, 1)$ and $v = (1, 1, 1, -1)$ in $\mathbb{R}^4$, where you also find the angle.

Answer. (i) $\langle u, v\rangle = 8$, $\|u\| = \|v\| = 3$, $\cos\theta = \frac{8}{9}$. (ii) $\langle u, v\rangle = 2$, $\|u\| = \|v\| = 2$, $\cos\theta = \frac{1}{2}$, $\theta = 60^\circ$.

Exercise 7.10 (computational) In $\mathbb{C}^2$ take $u = (1, i)$ and $v = (2, 1)$. Use the complex polarization identity to recover $\langle u, v\rangle$ from the four norms $\|v + \alpha u\|^2$ for $\alpha \in \{1, -1, i, -i\}$. Check your result directly.

Answer. $\langle u, v\rangle = 2 - i$; the four squared norms are $11, 3, 5, 9$ in the order $1, -1, i, -i$.

Hint. First compute the four vectors $v + \alpha u$ and their squared norms; the identity then builds $\langle u, v\rangle$ from these four numbers. Check the result against the formula $\overline{u_1} v_1 + \overline{u_2} v_2$.

Exercise 7.11 (proof) In a *real* inner product space, let $\|u\| = \|v\|$. Prove that $u+v$ and $u-v$ are orthogonal: the diagonals of a rhombus are perpendicular.

Hint. Expand $\langle u+v, u-v\rangle$ by linearity in both slots. Over $\mathbb{R}$ the inner product is symmetric, so the cross terms cancel and only the norms are left.

Exercise 7.12 ($\star$) Prove that the triangle inequality becomes the equality $\|u+v\| = \|u\|+\|v\|$ if and only if one of the vectors u, v is a multiple of the other with a nonnegative real multiplier.

Hint. Square the equality and compare it with the expansion of $\|u+v\|^2$. You reach the equality case of the Cauchy–Schwarz inequality; what remains is to pin down the phase of the multiplier.

Dirac notation

Exercise 7.13 (computational) Which of the following maps $\mathbb{C}^2 \to \mathbb{C}$ are linear functionals? Justify your answer: (i) $\varphi(v) = 3v_1 - iv_2$; (ii) $\varphi(v) = \overline{v_1}$; (iii) $\varphi(v) = v_1 v_2$; (iv) $\varphi(v) = \langle v, (1,i)\rangle$.

Answer. Only (i). (ii) and (iv): $\varphi(ie_1) = -i \neq i\,\varphi(e_1)$; (iii): $\varphi(2v) = 4\varphi(v)$ for $v = (1,1)$.

Hint. A linear functional satisfies $\varphi(au+bw) = a\varphi(u) + b\varphi(w)$ for all *complex* a, b. Try $a = i$ and $a = 2$.

Exercise 7.14 (computational)

(i) Let $|0\rangle, |1\rangle$ be an orthonormal basis, $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, and $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$. Find the outcome probabilities of a measurement in the basis $|0\rangle, |1\rangle$, and $\langle\sigma_z\rangle$.

(ii) Answer the same questions for $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$.

(iii) Take $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + i|1\rangle\big)$ and $|+\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$. Find the probabilities of the outcomes $|0\rangle$ and $|1\rangle$, the amplitude $\langle + \mid \psi\rangle$, and the probability $|\langle + \mid \psi\rangle|^2$.

Answer. (i) Probabilities $\frac{9}{25}$ and $\frac{16}{25}$; $\langle\sigma_z\rangle = -\frac{7}{25}$. (ii) Probabilities $\frac{1}{4}$ and $\frac{3}{4}$; $\langle\sigma_z\rangle = -\frac{1}{2}$. (iii) Probabilities $\frac{1}{2}$ and $\frac{1}{2}$; $\langle + \mid \psi\rangle = \frac{1+i}{2}$, $|\langle + \mid \psi\rangle|^2 = \frac{1}{2}$.

Exercise 7.15 (computational) In $\mathbb{C}^2$ take $|0\rangle = (1,0)$, $|1\rangle = (0,1)$, and $|\psi\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)$. Write down the matrix of the outer product $|\psi\rangle\langle\psi|$ in the basis $|0\rangle, |1\rangle$.

Answer. $|\psi\rangle\langle\psi| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Exercise 7.16 (proof) Let ϕ, ψ be nonzero and $T = |\psi\rangle\langle\phi|$. Prove that $T^2 = \langle\phi \mid \psi\rangle\, T$, and find when $T^2 = T$.

Answer. $T^2 = T$ exactly when $\langle\phi \mid \psi\rangle = 1$.

Hint. Apply T^2 to an arbitrary vector $|\chi\rangle$; the number $\langle\phi \mid \psi\rangle$ comes out of the middle.